



Machine Learning Tutorial

k-Nearest Neighbours Classification and the Effect of the
Hyperparameter k on Decision Boundaries

Module: Machine Learning and Neural Networks

Course: MSc Data Science

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GitHub Link:

https://github.com/sohailahmed01/ML_assignment_KNN_decision_boundary.git

Dataset Link: <https://archive.ics.uci.edu/dataset/53/iris>

Submission Type: Tutorial (PDF) + Jupyter Notebook + GitHub Repository

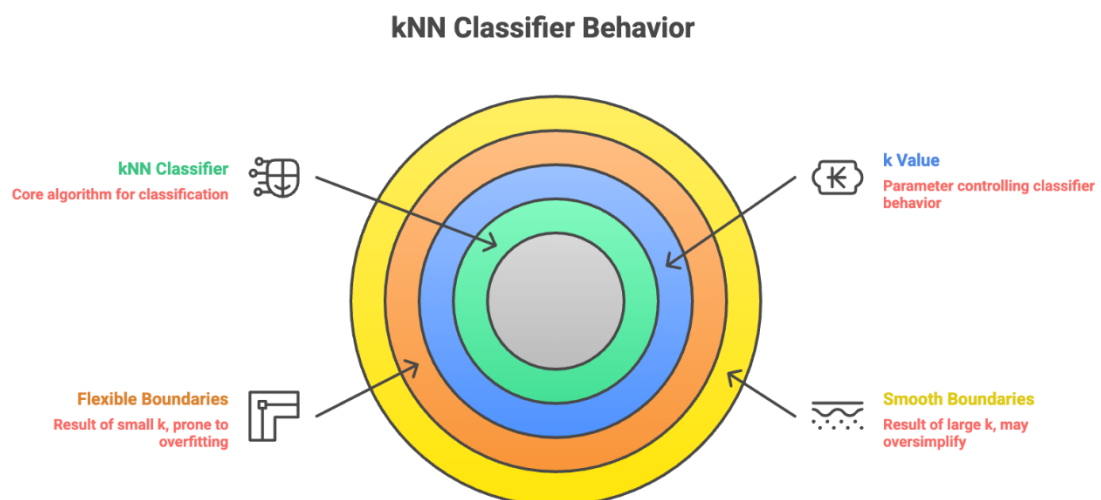
Abstract

This tutorial examines the behaviour of the k-Nearest Neighbours (kNN) algorithm with a specific focus on how the hyperparameter k influences decision boundaries, model flexibility, and classification accuracy. Although kNN is conceptually simple, variations in k can produce fundamentally different classification surfaces and generalisation performance. The tutorial demonstrates these effects on synthetic two-dimensional moon-shaped data and on the Iris dataset. Cross-validation is used to identify an optimal k , and the observed patterns are interpreted in the context of the bias–variance trade-off. All experiments are fully reproducible through the accompanying Jupyter notebook.

1. Introduction

k-Nearest Neighbours (kNN) is a classical supervised learning algorithm that assigns class labels based on the majority vote among the k closest training instances. Despite its simplicity, kNN remains highly competitive for many classification tasks, particularly those involving well-separated clusters and low-dimensional spaces. However, the value of k fundamentally controls the behaviour of the classifier. Small values produce highly flexible boundaries which may lead to overfitting, whereas large values generate smoother, biased boundaries that may oversimplify underlying class structures.

This tutorial investigates how different values of k influence decision boundaries and model performance. Two datasets are used: a synthetic noisy moons dataset to visually illustrate boundary changes, and the Iris dataset to demonstrate the practical selection of k using cross-validation. All results align with reproducible machine learning principles and the practices recommended in the academic literature ((Cover and Hart, 1967; Duda, Hart and Stork, 2001)).



2. Theoretical Background

kNN is a non-parametric classifier, meaning it does not learn an explicit global model. Instead, predictions are made solely from local neighbourhood relationships in the feature space. For an unseen observation, the Euclidean distance to each training sample is typically computed, and the most common class among the k nearest neighbours is returned as the predicted label.

The choice of k determines the model's flexibility. When $k = 1$, the classifier exactly memorises the training data, leading to highly complex and unstable decision boundaries. Larger values of k progressively smooth these boundaries, reducing variance while increasing bias. This behaviour reflects the classical bias–variance trade-off (Hastie, Tibshirani and Friedman, 2009). Selecting k via cross-validation is therefore essential for achieving optimal generalisation.

Feature scaling is necessary when working with kNN because distance metrics are highly sensitive to differences in feature magnitudes, particularly in datasets with heterogeneous units.

3. Synthetic Data Experiment: Moons Dataset

A two-dimensional moons dataset with 300 samples and moderate noise was generated to illustrate the effect of k on the geometry of the decision boundary. This dataset contains two interlocking semicircular clusters with nonlinear separability, making it suitable for observing changes in boundary complexity.

3.1 Visualisation of the Raw Data

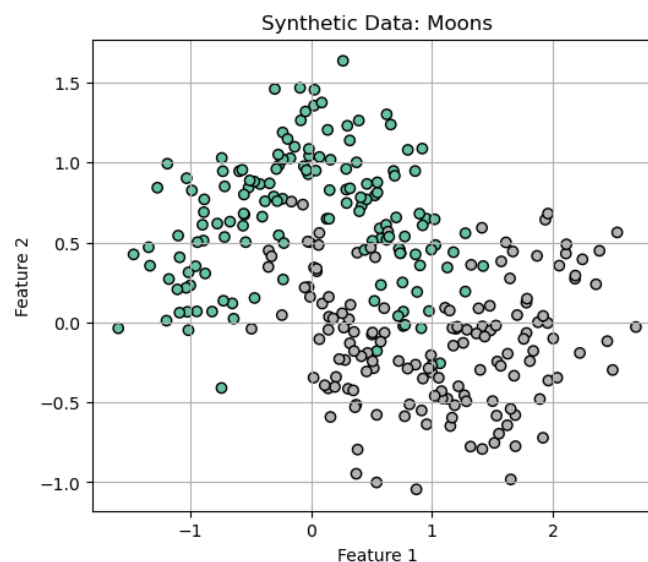


Figure 1 Synthetic Moons Dataset.

The scatter plot reveals two crescent-shaped clusters with overlapping regions due to added noise. The non-linear structure requires a classifier capable of flexible local modelling.

3.2 Effect of k on Decision Boundaries

Values of $k = 1, 3, 5, 15$, and 50 were used to train kNN classifiers. The resulting boundaries exhibit progressive smoothing.

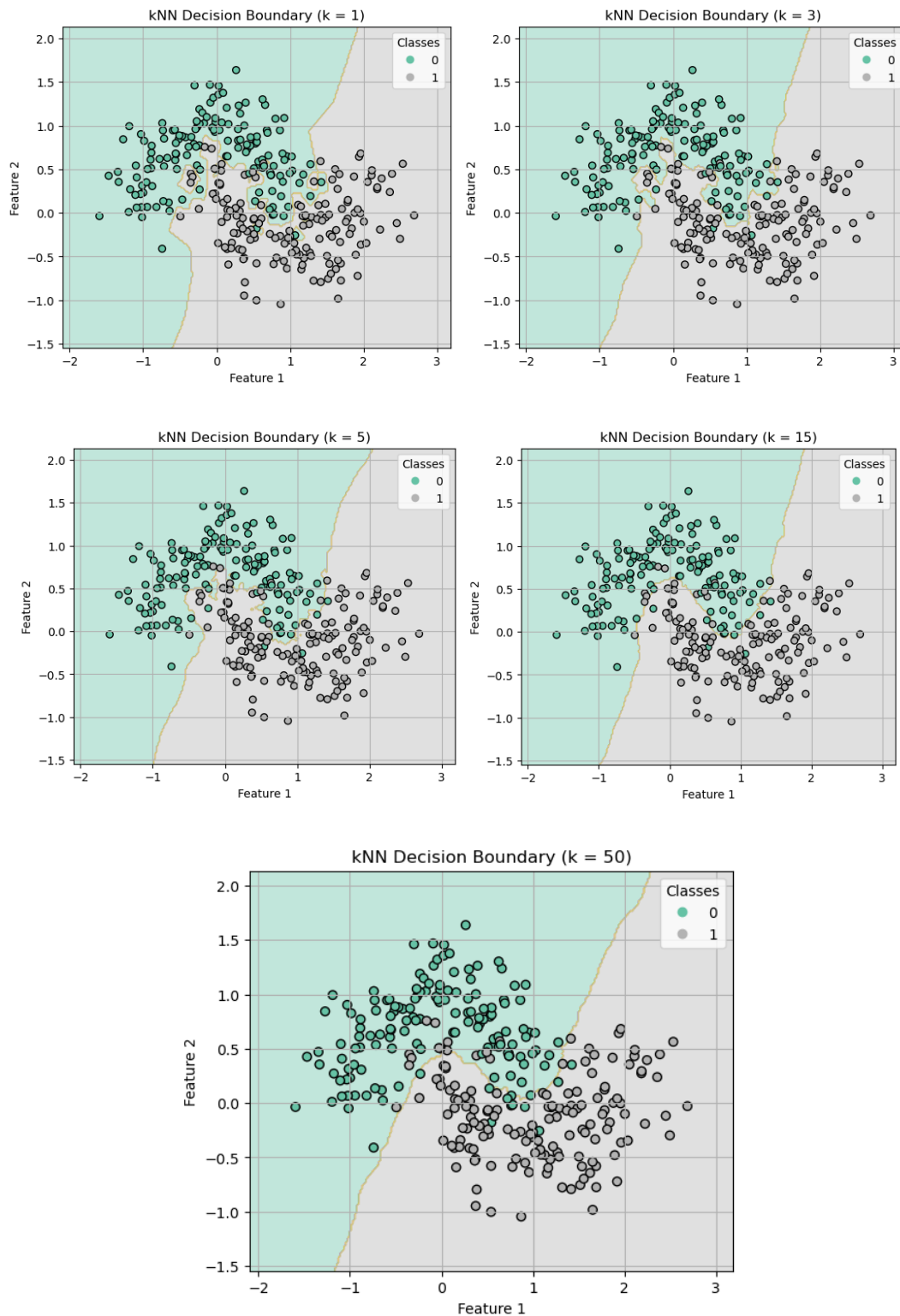


Figure 2 Decision Boundary for $k = 1, 3, 5, 15, 50$.

The classifier with $k = 1$ produces a highly irregular decision surface that tracks noise fluctuations in the training set. As k increases, the boundary becomes smoother, leading to a monotonic decrease in training accuracy and a gradual reduction in test accuracy. The most significant value, $k = 50$, generates an excessively smooth boundary that fails to respect the natural shape of the clusters.

3.3 Quantitative Evaluation on Train and Test Splits

The table below summarises training and testing accuracy across the considered values of k .

Table 1. Accuracy comparison for different values of k on the moons dataset

k	Training Accuracy	Testing Accuracy
1	1.00	0.9556
3	0.9476	0.9556
5	0.9333	0.9444
15	0.9238	0.9444
50	0.9048	0.9222

The results show that the training accuracy decreases steadily as k increases. Testing accuracy is highest for $k = 1$ and $k = 3$, after which it declines due to over-smoothing. These findings are consistent with theoretical expectations and demonstrate the importance of model selection.

4. Iris Dataset Experiment

The Iris dataset contains 150 samples evenly distributed across three species: setosa, versicolor and virginica. Each sample includes four numerical features describing the plant's sepal and petal morphology (FISHER, 1936). A stratified train-test split was used to maintain class proportions.

4.1 Exploratory Analysis

The class distribution is perfectly balanced.

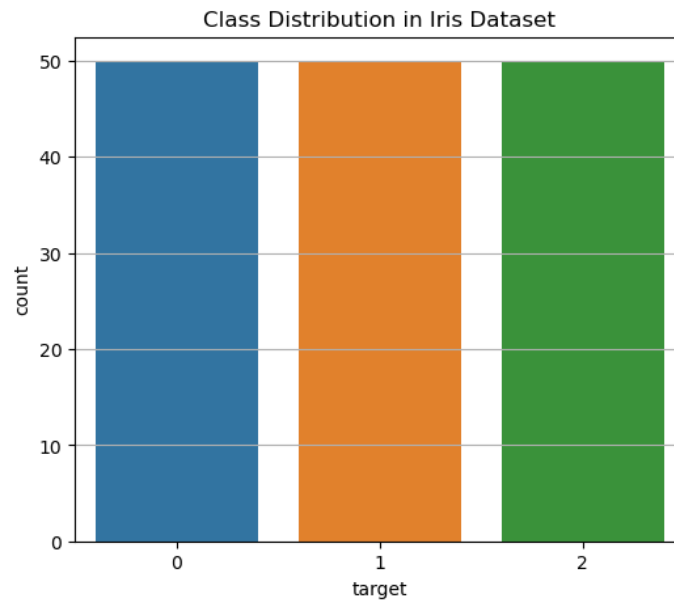


Figure 3 Class Distribution in Iris Dataset.

A two-dimensional projection using petal length and petal width shows clear separation of setosa but partial overlap between versicolor and virginica.

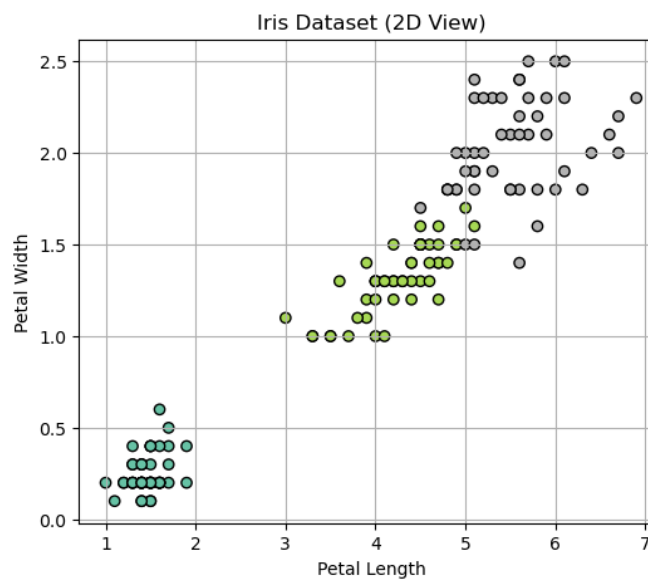


Figure 4 Iris Dataset (2D View).

4.2 Hyperparameter Selection via Cross-Validation

A kNN classifier with standardised features was evaluated using five-fold cross-validation for odd values of k between 1 and 31. The mean cross-validation accuracy peaked at approximately 0.966 for several values less than or equal to $k = 15$.

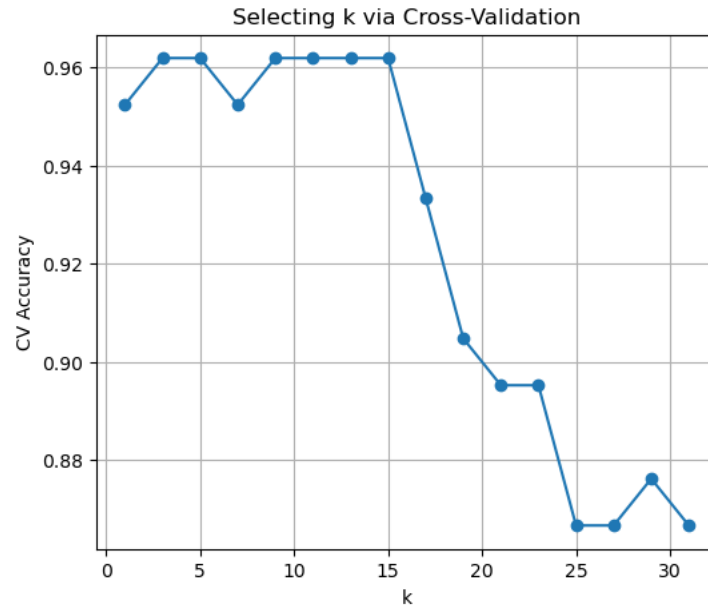


Figure 5 Cross-Validation Accuracy vs k .

The curve demonstrates high performance for moderate values of k , followed by a notable accuracy decline once k exceeds 15. This behaviour captures the transition from low-bias, high-variance regimes to oversmoothed, high-bias classifiers.

The optimal value selected by cross-validation was consistent with the top of this plateau.

4.3 Model Evaluation on the Test Set

Using the optimal model, a test accuracy of **0.9111** was obtained. The classification report is summarised below.

Table 2. Classification metrics for the Iris dataset

Class	Precision	Recall	F1-score	Support
0	1.00	1.00	1.00	15
1	0.79	1.00	0.88	15
2	1.00	0.73	0.85	15

Although classes 0 and 1 are correctly identified with high recall, class 2 exhibits reduced recall due to overlap with class 1 in the feature space. These outcomes are well-aligned with previous analyses of the Iris dataset in the literature (Duda et al., 2001).

The corresponding confusion matrix is shown below.

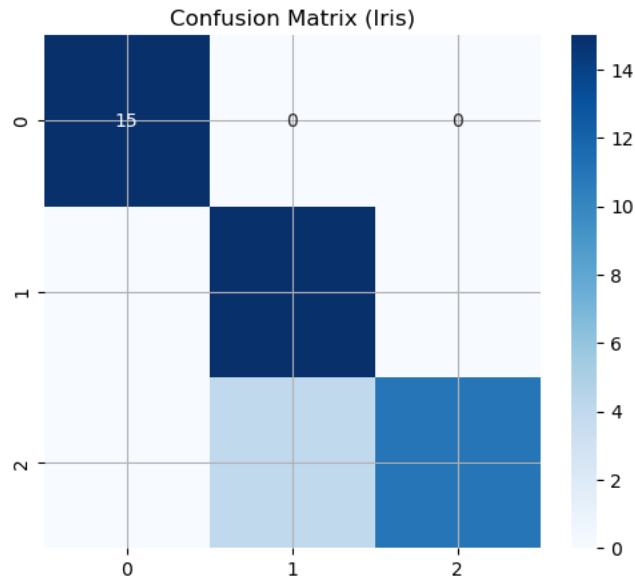
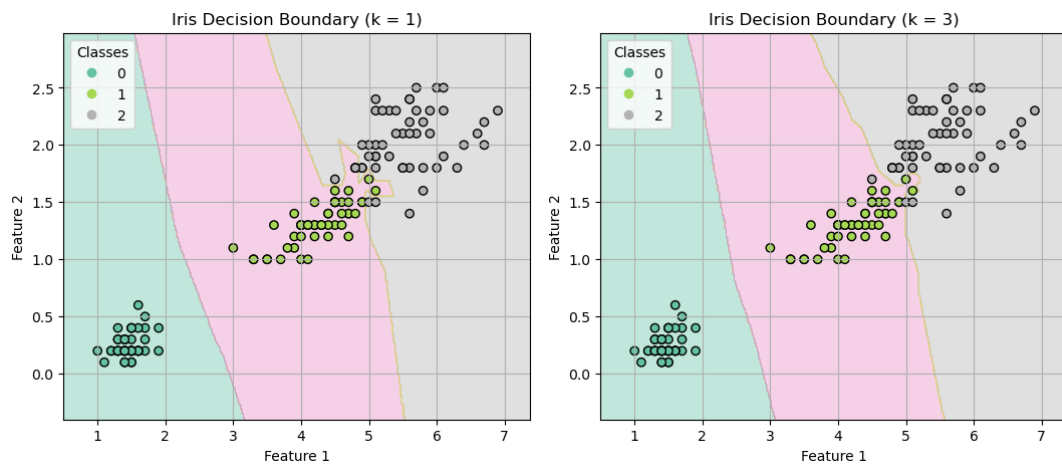


Figure 6 Confusion Matrix.

The matrix visually confirms that misclassifications occur primarily between classes 1 and 2, while class 0 remains perfectly separable.

4.4 Visualisation of Decision Boundaries in Two Dimensions

To further interpret the effect of k , a reduced feature space consisting of petal length and petal width was used. Decision boundaries were plotted for $k = 1$, the cross-validated k , and a large value such as $k = 25$.



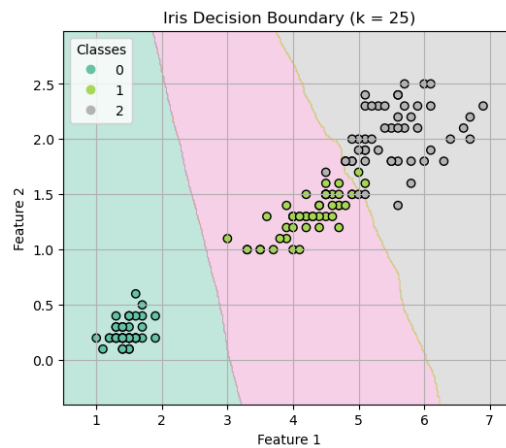


Figure 7 Iris Decision Boundary ($k = 1$, $k=3$, $k=25$).

Although the actual classification takes place in four dimensions, such projections yield meaningful two-dimensional comparisons. The boundary for $k=1$ suffers from some locality problems. Conversely, $k=25$ creates too simple boundaries, which hide the difference between the versicolor and virginica species.

5. Discussion

In both experiments, the importance of k , as the controlling factor for kNN activity, becomes clear. Low values are associated with high flexibility, which may result in over-fitting and non-linear decision boundaries. As k increases, the model becomes less flexible but prone to over-smoothing, especially for non-linearly separable boundaries.

Cross-validation curves for the Iris data follow the same trend. The range where high levels of accuracy are observed up until $k = 15$ indicates that multiple values of k are optimal for generalisation. This trend, deteriorating after that point, resembles the effects of underfitting. This classification report indicates that the error distribution between the versicolor and virginica classes consists of the most number of errors, owing to the overlap present between the two classes. This provides the significance for the evaluation of the accuracy for each class.

6. Ethical Considerations in Machine Learning

While the kNN (k-Nearest Neighbors) algorithm is fairly straightforward and straightforward, it also poses numerous ethical concerns. kNN makes predictions based upon the characteristics of surrounding points, so any biases or inequalities that exist within the training data will propagate through to the model predictions. This is especially true in sensitive fields such as health care, hiring and finance where many classification algorithms are used. Additionally, using distance-based measures to classify can lead to prioritising certain features over others that reflect social or historical injustices if the data sets were not well-thought-out. To reduce the ethical risks associated with these systems, kNN practitioners should be proactive about ensuring their data sets are representative, their data pre-processing steps do not increase bias and that they continuously evaluate the performance of their models across all demographic groups. Another aspect of ethical deployment is transparency; because kNN's method of determining the classification it produces can be explained using the neighbours involved, being able to explain how kNN makes its predictions may improve fairness and accountability when utilised in the real world.

7. Reproducibility Note

All figures, tables, and numerical values shown and discussed in the tutorial are produced inside the accompanying Jupyter Notebook. This provides the entire code for the data preparation, model development, and evaluation. All values discussed and shown are completely reproducible inside any standard Python environment that supports scikit-learn, NumPy, pandas, and matplotlib.

8. Conclusion

In this tutorial, the impact of the hyperparameter k on the characteristics of kNN classifier algorithms based on boundary complexity, bias and variance, and accuracy levels has been illustrated. From the experiments conducted, it was confirmed that the accuracy levels of the kNN classifier algorithm are optimal for values of k that are not extremal. Cross-validation provides an effective technique of choosing an optimal k , and the graphical representation of the decision boundary provides an intuitive validation of the kNN classifier performance.

How to Run This Notebook

To reproduce all results shown in the tutorial:

1. Install dependencies

```
pip install -r requirements.txt
```

2. Open the notebook

```
jupyter notebook knn_decision_boundaries.ipynb
```

3. Run all cells

Use: **Kernel** → **Restart & Run All**

All figures will be generated inside the figures/ directory.
Random seeds ensure full reproducibility across environments.

9. Accessibility Statement

To ensure this tutorial is accessible to a wide audience, several measures were incorporated during its preparation. All figures use colourblind-safe palettes and include descriptive captions to support readers with visual impairments. The document layout is compatible with standard screen-reader software, and images contain meaningful descriptions that do not rely solely on colour to convey information. The accompanying notebook reproduces all results in a clear and structured manner, allowing users with different sensory needs or learning styles to access the content effectively. Overall, the tutorial aims to provide an inclusive learning experience through thoughtful visual design and clear explanatory text.

10. References

Cover, T.M. and Hart, P.E. (1967) 'Nearest neighbor pattern classification', *IEEE Transactions on Information Theory*, 13(1), pp. 21–27. doi:10.1109/TIT.1967.1053964.

Duda, R.O., Hart, P.E. and Stork, D.G. (2001) 'Chapter 9 - Algorithm-Independent Machine Learning', *Pattern Classification*, p. 680. Available at: <https://www.wiley.com/en-kr/Pattern+Classification%2C+2nd+Edition-p-9780471056690> (Accessed: 4 December 2025).

FISHER, R.A. (1936) 'THE USE OF MULTIPLE MEASUREMENTS IN TAXONOMIC PROBLEMS', *Annals of Eugenics*, 7(2), pp. 179–188. doi:10.1111/J.1469-1809.1936.TB02137.X.

Hastie, T., Tibshirani, R. and Friedman, J. (2009) 'Introduction', pp. 1–8. doi:10.1007/978-0-387-84858-7_1.